PROJECT DESIGN PHASE-II

Technology Stack Architecture

Developer: **Bhargavasai** • Credit Card Fraud Detection System

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PRESENTATION LAYER

Frontend UI Web Interface

`HTML5, CSS3, JavaScript`

Builds a clean, fully responsive input control panel dashboard. It enables end-users and compliance evaluators to smoothly type numerical parameters (Time, V1-V28 vectors, Amount) and visually displays high-impact alert indicators ("Normal" vs "Fraud Detected") instantly upon inference completion.

LOGIC LAYER

Backend API Controller Application

`Python, Flask Framework`

Serves as the high-speed structural communication router core. It intercepts active HTTP POST data payloads from the UI client, executes strict type parsing validations, routes arrays directly through data normalizers, and interfaces dynamically with local serialized machine learning prediction engines.

INTELLIGENCE LAYER

Machine Learning Architecture

`Scikit-learn, XGBoost, Pandas, NumPy, Joblib`

The core calculation pipeline engine of the ecosystem. Implements structured data preprocessing scaling configurations via StandardScaler. Employs advanced optimized gradient boosting trees (XGBoost) alongside Random Forest configurations to pinpoint fraudulent transactional anomalies with high precision, loaded instantly via single-cycle Joblib methods.

INFRASTRUCTURE LAYER

Version Control & Cloud Deployment

`GitHub, Render Cloud Services`

Manages production pipelines and development cycles. Code repositories are maintained with strict version tracking trees inside GitHub. Continuous integration routines push verified updates directly onto Render hosting frameworks, serving active live evaluation pathways via production-ready public endpoints.
